METRIC

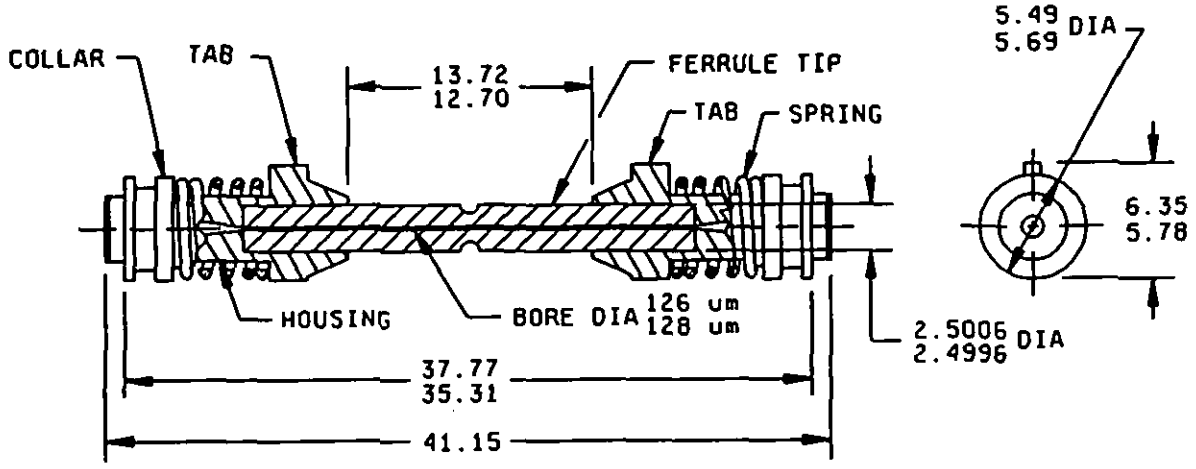
MIL-S-24623/4B
21 JUNE 1995
SUPERSEDING
MIL-S-24623/4A
10 June 1991

MILITARY SPECIFICATION SHEET

SPLICE, FIBER OPTIC, HOUSING, ROTARY MECHANICAL, FIBER

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: DOD-S-24623.



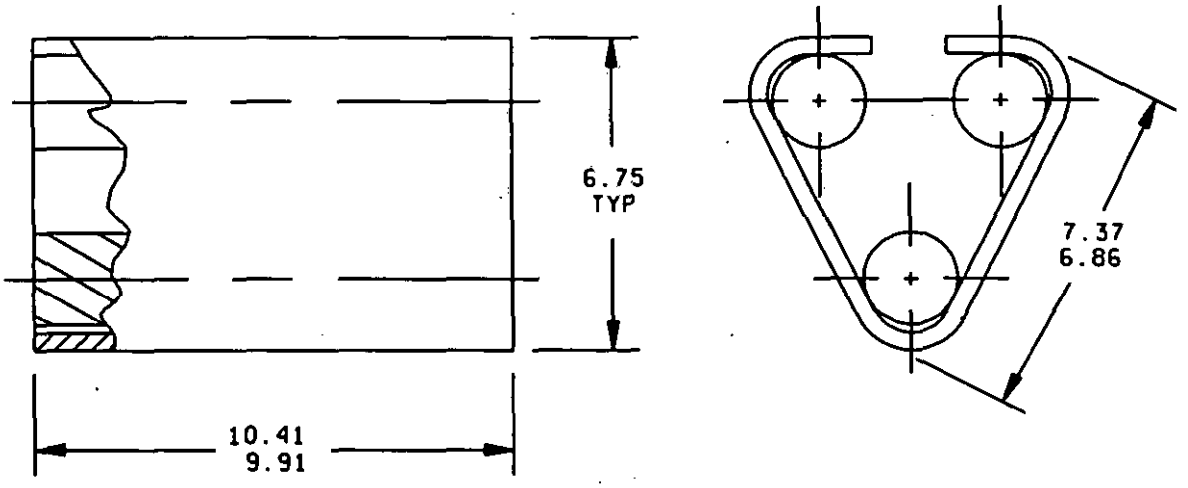
mm	Inches
2.4996	.09841
2.5006	.09845
5.49	.216
5.69	.224
5.78	.228
5.84	.230
6.35	.250
12.70	.500
13.72	.540
35.31	1.390
37.77	1.487
41.15	1.620

NOTES:
1. All dimensions are in millimeters.
2. Inch equivalents are given for general information only.

FIGURE 1. Rotary splice ferrule - M24623/04-04.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:26Z
Check the source to verify that this is the current version before use.

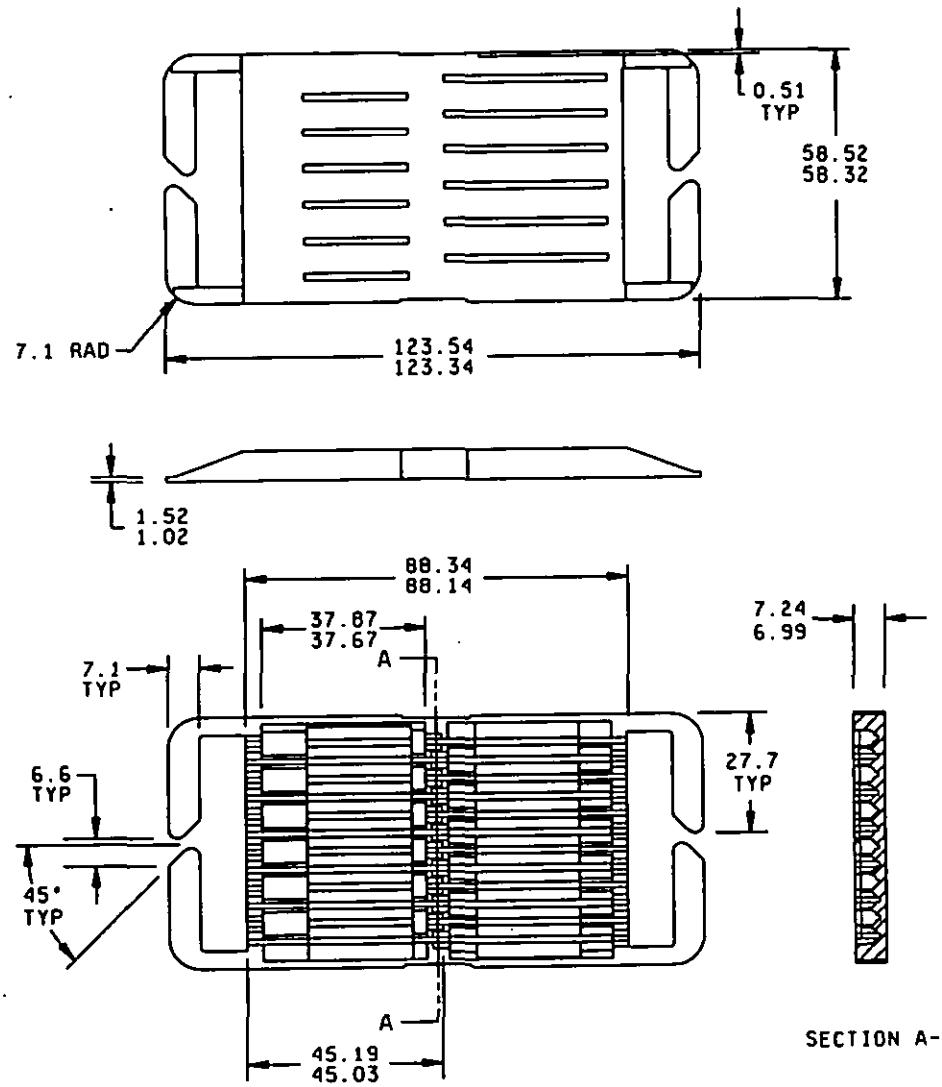


mm	Inches
6.75	.266
6.86	.270
7.37	.290
9.91	.390
10.41	.410

NOTES:
1. All dimensions are in millimeters.
2. Inch equivalents are given for general information only.

FIGURE 2. Rotary splice triangular sleeve - M24623/04-05.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:26Z
Check the source to verify that this is the current version before use.

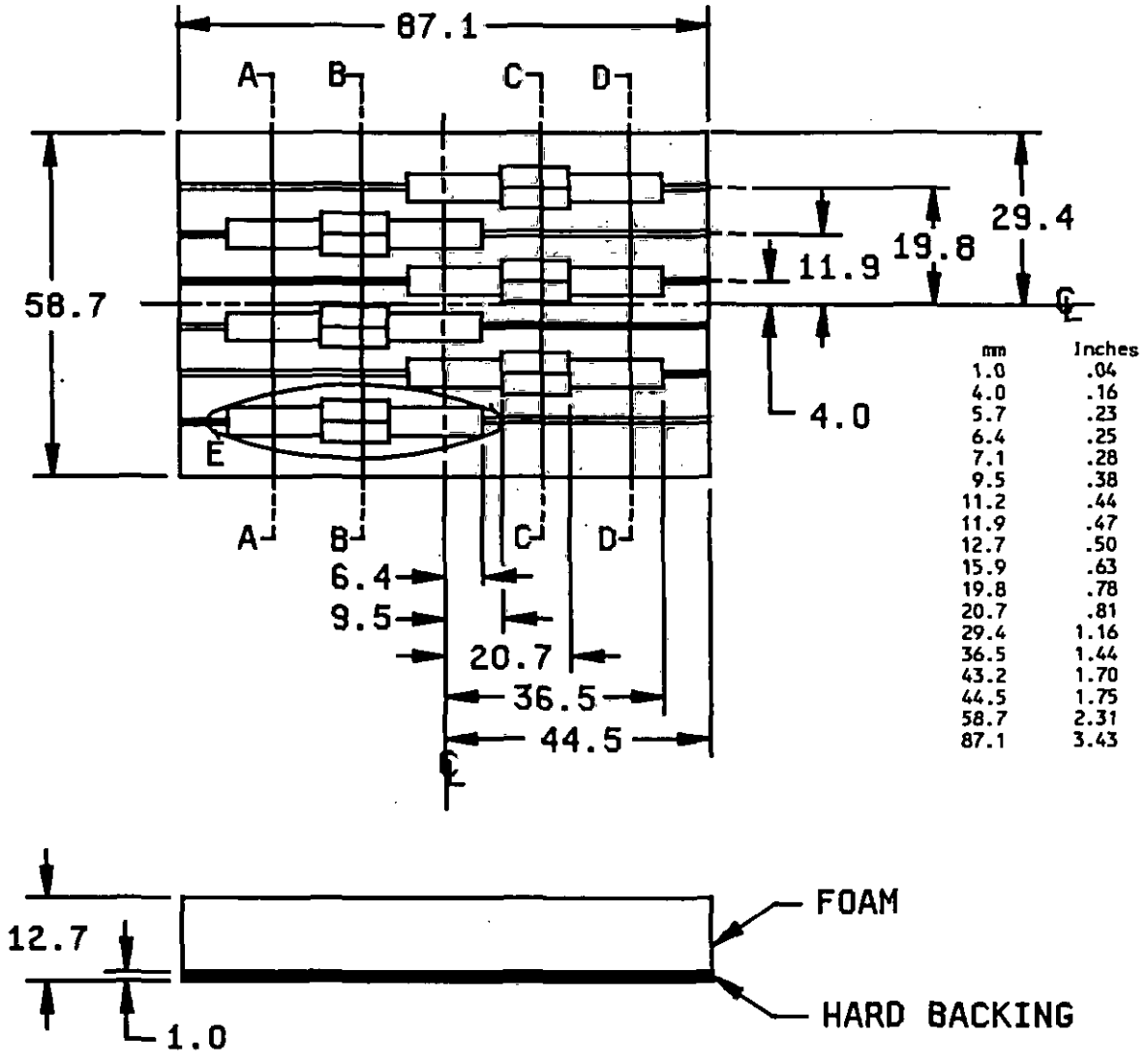


mm	Inches
.51	.020
1.02	.040
1.52	.060
6.6	.26
6.99	.275
7.1	.28
7.24	.285
27.7	1.09
37.67	1.483
37.87	1.490
45.03	1.773
45.19	1.779
58.32	2.296
58.52	2.304
88.14	3.470
88.34	3.478
123.34	4.856
123.54	4.864

NOTES:
1. All dimensions are in millimeters.
2. Inch equivalents are given for general information only.

FIGURE 3. Rotary splice tray - M24623/04-02.

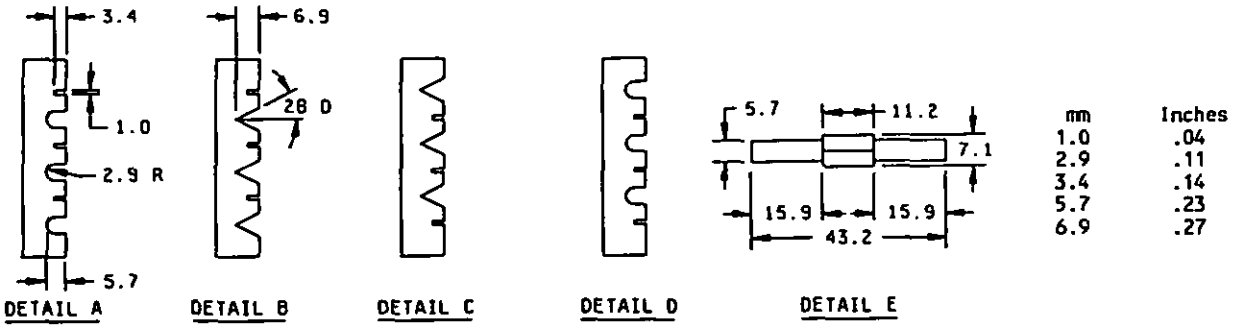
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:26Z
Check the source to verify that this is the current version before use.



NOTES:
1. All dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .5 (.02) for 1 place decimals.

FIGURE 4. Rotary splice tray (shock hardened) - M24623/04-03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:26Z
Check the source to verify that this is the current version before use.



NOTES:
1. All dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.5 mm (.02 inch) for 1 place decimals.
4. Unless otherwise specified, tolerance on all angles is ±2°.

FIGURE 4. Rotary splice tray (shock hardened) - M24623/04-03 - Continued.

REQUIREMENTS:

Design and construction: See figures 1, 2, 3, and 4.

Fiber style: Both single-mode and multimode.

Splice type: Fiber splice.

Weight: Rotary splice (with triangular sleeve): ≤ 3 grams.

Rotary splice tray: ≤ Less than or equal to 14 grams.

Ferrule material: Glass or ceramic.

Exterior parts: Metallic exterior parts of the assembled splice shall have a passivated finish compatible with external coatings or platings of the type and color specified. The alignment sleeve may be beryllium copper.

Nuclear radiation: Not applicable.

Freezing water: Not applicable.

Fluid immersion: Not applicable.

Sand and dust: Not applicable.

Crosstalk: Not applicable.

Salt spray: Not applicable.

Water pressure: Not applicable.

Twist: Not applicable.

Cable seal flexing: Not applicable.

Axial compression loading: Not applicable.

Cable pull out: Not applicable.

Crush: Not applicable.

Flammability: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:26Z
Check the source to verify that this is the current version before use.

Temperature range: 1.

Qualification:

M24623/04-01 and -04 - The following tests apply: Visual and mechanical inspections, mass, size, workmanship, insertion loss, return loss, crosstalk, ambient light susceptibility, impact, operating temperature, thermal shock, fiber dynamic strength, temperature/humidity, fungus, temperature life, mechanical shock, and vibration.

M24623/04-02 - The following tests shall apply: Visual and mechanical inspections, mass, size, workmanship, operating temperature, thermal shock, fungus, mechanical shock, and vibration. For the thermal shock, operating temperature, mechanical shock, and vibration tests, the sample shall be tested using two splices in accordance with figure 1. When samples are tested using splices, one splice shall be placed in an edge location and one splice shall be placed in a random location in each test sample. During the mechanical shock test the optical splice may have discontinuities greater than 0.3 dB.

M24623/04-03 - The following tests apply: Visual and mechanical inspections, mass, size, workmanship, operating temperature, thermal shock, fungus, mechanical shock, and vibration. For the thermal shock, operating temperature, mechanical shock, and vibration tests, the sample shall be tested using two splices in accordance with figure 1. When samples are tested using splices, one splice shall be placed in an edge location and one splice shall be placed in a random location in each test sample.

Quality conformance (groups A, B, and C):

M24623/04-01 and -04 - Group B testing shall not be performed. The following tests apply in groups A and C: Visual and mechanical inspections, mass, size, workmanship, insertion loss, return loss, crosstalk, ambient light susceptibility, impact, operating temperature, thermal shock, fiber dynamic strength, temperature/humidity, temperature life, mechanical shock, and vibration. For the size inspections of group A only the ferrule external diameter and the housing-to-housing separation dimensions shall be tested.

M24623/04-02 - Group B testing shall not be performed. The following tests apply in groups A and C: Visual and mechanical inspections, mass, size, workmanship, operating temperature, thermal shock, mechanical shock, and vibration. For the size inspections of group A, the inspection may be conducted on a sampled basis. For the thermal shock, operating temperature, mechanical shock, and vibration tests, the sample shall be tested using two splices in accordance with figure 1. When samples are tested using splices, one splice shall be placed in an edge location and one splice shall be placed in a random location in each test sample. During the mechanical shock test the optical splice may have discontinuities greater than 0.3 dB.

M24623/04-03 - The following tests apply: Visual and mechanical inspections, mass, size, workmanship, operating temperature, thermal shock, mechanical shock, and vibration. For the size inspections of group A, the inspection may be conducted on a sampled basis. For the thermal shock, operating temperature, mechanical shock, and vibration tests, the sample shall be tested using two splices in accordance with figure 1. When samples are tested using splices, one splice shall be placed in an edge location and one splice shall be placed in a random location in each test sample.

Tools required: Refer to NAVSEA drawing 6872812.

Adhesive: AT&T Part or Identifying number (PIN) D-181729 or equivalent.

Index matching material: AT&T PIN AT-8955 or equivalent.

PIN:

M24623/04-01 Rotary splice ferrule and triangular sleeve. (see figures 1 and 2)

M24623/04-02 Rotary splice tray (non shock hardened). (see figure 3)

M24623/04-03 Rotary splice tray (shock hardened). (see figure 4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:26Z
Check the source to verify that this is the current version before use.

M24623/04-04 Rotary splice ferrule. (see figure 1)

M24623/04-05 Triangular sleeve. (see figure 2)

TABLE I. Supersession data.

PIN	Superseding
M24623/04-01	M24623/04-01* 1/
M24623/04-02	M24623/04-02
M24623/04-03	None
M24623/04-04 2/	None
M24623/04-05 2/	None

1/ * is any letter from A to L.
2/ For replacement purposes, the Government will stock and issue M24623/04-01.

Intended use: The splices and shock hardened splice tray are intended for use in all applications inside protective enclosures. The non-hardened splice tray is intended for use in protective enclosures that will not be exposed to shock or in shock isolated protective enclosures.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - EC, YD
Air Force - 17, 19, 80, 99
DLA - ES

Preparing activity:
Navy - SH

Agent:
DLA - ES

(Project 6060-0110-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:26Z
Check the source to verify that this is the current version before use.